

$$x + y = 3$$

$$x - y = 1$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Add together to eliminate y

$$x + x = 3 + 1$$

$$2x = 4$$

$$x = 2$$

Plug in $x = 2$

$$2 + y = 3 \quad y = 1$$

Use coefficients to make a matrix

$$2x + 3y = 0$$

$$4x + 5y = 0$$

$$x + \frac{3}{2}y = 0$$

$$\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$$

$$\begin{aligned} 4x + 5y \\ - 4(x + \frac{3}{2}y) &= 0 \\ -y &= 0 \end{aligned}$$

Eliminate by adding
multiple of row

Row reduction

$$\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3/2 \\ 4 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3/2 \\ 0 & -1 \end{pmatrix}$$

$$\begin{aligned} (4 \ 5) - 4(1 \ 3/2) \\ = (0 \ 5-6) = (0 \ -1) \end{aligned}$$

$$X = 0$$

1 equation 1 unknown

Have unique solution

$$X = 0 \quad 2 \text{ eqns } 1 \text{ unknown}$$

$X = 1$ No sol's. Sol true all eqns

$$X = 0$$

$$2X = 0$$

There is a solution

$$X + y = 0 \quad 1 \text{ eqn } 2 \text{ unknowns}$$

Infinite solutions $y = -X$

$$X + y + z = 0 \Rightarrow Z = -\underset{\uparrow}{X} - \underset{\uparrow}{y}$$

Parameters

$$X + y + z = 0$$

$$X - y + z = 0$$

Constraint
 $2y = 0$
 $y = 0$

$$Z = -X$$

m equations

n variables

($M \times N$ matrix
m rows
n columns)

IF $m = n$

We expect a unique solution
(No guarantee)

$m = \#$ Constraints

$n = \#$ Free variables

IF $m > n$

$$x = 0$$

$$x = 1$$

or

$$x = 0$$

$$2x = 0$$

expect no solutions

(No guarantee)

IF ~~$m > n$~~ $m < n$

expect ∞ sol's.

$$x + y = 0$$

(No guarantee)

Usually not obvious
When redundant or not

$$x + 2y + 3z = 0$$

$$-2x - y + 5z = 4$$

$$-x + y + 8z = 4$$



Redundant $\textcircled{0} + \textcircled{0} = \textcircled{0}$

Could have been less obvious

$$2 \textcircled{0} + 5/4 \textcircled{0} = \textcircled{0}$$

$$-3/2 x + 3/4 y + 37/4 z = 5$$

